

Optics for High-Speed PIV

Field of view, magnification, f-number, diffraction
and particle image size

A practical reference covering the optical chain from sensor pixel to velocity vector. All numbers are worked for a representative system: a 2560×1664 , $9.27 \mu\text{m}$ pixel, Super 35 high-speed camera (Phantom T2540 class) paired with a 1 kHz dual-cavity Nd:YLF laser at 527 nm. The formulae are general; substitute your own hardware.

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Parameter	Value	Consequence
Sensor resolution	2560 × 1664 (4.26 Mpx)	Sets pixels per frame
Pixel pitch, p	9.27 μm (18.54 μm binned)	Sets Nyquist limit, diffraction target
Sensor size	23.7 × 15.4 mm (28.2 mm diag.)	Super 35; sets field of view
Wavelength, λ	527 nm (Nd:YLF, doubled)	Sets diffraction scale
Sensor type	BSI CMOS, global shutter	Wide acceptance angle; no motion skew
Bit depth	12 bit	6.39 MB per frame
Throughput	25 Gpx/s	Sets all frame rates
Quantum efficiency	85.4% mono at 532 nm	Measured at the laser line
Dynamic range	52.0 dB (8.6 stops)	Expose carefully; little headroom
RAM buffer	64 / 128 / 256 GB	Sets record duration
Straddle time	364 ns minimum	Sets shortest achievable Δt
Lens mount	F, EF, PL, C, M42	C-mount will not cover 28.2 mm

Table 1: Reference system. The first four rows drive everything else.

1 The reference system

Every number in this document follows from the four highlighted rows in Table 1.

A note on the C-mount option. Mount type does not guarantee image circle coverage. C-mount lenses are designed for 2/3 inch to 1 inch formats (11–16 mm diagonal) and will vignette hard on a 28.2 mm diagonal sensor. They are usable only with a cropped region of interest of roughly 1000×650 or smaller. F-mount, EF and PL all cover the full sensor.

2 Throughput and frame rate

Pixel throughput, quoted in gigapixels per second ($1 \text{ Gpx/s} = 10^9$ pixel values per second), is the single number from which every frame rate in a high-speed camera datasheet derives. It describes how fast the sensor readout, ADC array and memory interface move pixel values end to end.

$$\text{fps} = \frac{\text{throughput}}{\text{pixels per frame}} \quad (1)$$

Resolution	Pixels/frame	25 Gpx/s ÷ px	Datasheet
2560 × 1664	4,259,840	5,870	5,840
2048 × 1152	2,359,296	10,600	8,440
1280 × 832 (binned)	1,064,960	23,470	23,250
2560 × 256	655,360	38,150	37,590
2560 × 32	81,920	305,200	277,770

Table 2: Equation (1) against published values. Shortfalls are row-timing and readout overhead, which grow proportionally as the region of interest shrinks.

Converting throughput to a data rate makes the architecture obvious: $25 \text{ Gpx/s} \times 12 \text{ bit} = 300 \text{ Gbit/s} = 37.5 \text{ GB/s}$ sustained. No storage device accepts that, which is why high-speed cameras record into a circular RAM buffer rather than to disk. A 256 GB buffer at full resolution and maximum frame rate holds about 6.9 seconds.

Reducing the region of interest is *not* a zoom. It reads a smaller window out of the same sensor at the same magnification, so the field of view shrinks. Choose focal length for the full sensor first, then crop to the shape the event actually occupies. This is why high-speed setups so often use wide letterbox regions — ballistics, droplets and crack fronts travel along one axis.

3 The light budget

The defining constraint of high-speed imaging is not speed but light. Maximum exposure per frame is the frame interval.

Frame rate	Max exposure	Light vs. 1/50 s
50 fps (cinema reference)	20,000 μ s	1 $\times$
2,000 fps	500 μ s	40 $\times$ less
5,840 fps	171 μ s	117 $\times$ less
37,590 fps	26.6 μ s	750 $\times$ less
277,770 fps	3.6 μ s	5,600 $\times$ less

Table 3: Exposure and relative light per frame.

This is the entire design logic behind large pixels (9.27 μ m, roughly five times the area of a typical DSLR pixel), back-side illumination and 85% quantum efficiency. All three exist to claw back photons. It is also why exposure being *independent of frame rate* matters on these cameras: you can shoot 1,000 fps with a 1 μ s exposure to freeze motion within a single frame, then wait out the rest of the interval.

4 Magnification and object-space resolution

Magnification is the ratio of image size on the sensor to real size on the object. It is dimensionless, and for a thin lens follows from $1/f = 1/s + 1/s'$:

$$m = \frac{f}{s - f} \quad (2)$$

where f is focal length and s the object-to-lens distance, measured to the front principal plane. The two quantities that matter in practice are the field of view and the object-space pixel — the real-world distance one pixel covers, sometimes called the scale factor or simply mm/pixel:

$$\text{FOV} = \frac{\text{sensor dimension}}{m}, \quad p_{\text{obj}} = \frac{p}{m} \quad (3)$$

Worked for a 100 mm lens at $s = 1000$ mm: $m = 100/900 = 0.111$, giving $p_{\text{obj}} = 9.27/0.111 = 83.4$ μ m and a field of view of 213×139 mm.

Setup	m	Object pixel	FOV width
50 mm @ 1.0 m	0.053	176 μ m	450 mm
90 mm @ 1.0 m	0.099	94 μ m	240 mm
100 mm @ 1.0 m	0.111	83 μ m	213 mm
180 mm @ 2.0 m	0.099	94 μ m	240 mm
180 mm @ 1.0 m	0.220	42 μ m	108 mm
100 mm @ 0.5 m	0.250	37 μ m	95 mm

Table 4: Rows 2 and 4 are optically equivalent: doubling focal length and doubling distance cancel.

4.1 Designing backwards

In practice you know the field of view you need and the standoff you are allowed, and you want the lens:

$$m = \frac{\text{sensor dimension}}{\text{FOV}_{\text{required}}}, \quad f = \frac{m s}{1 + m} \quad (4)$$

Example: to cover a 300 mm wide event from 1.5 m, $m = 23.7/300 = 0.079$, so $f = 0.079 \times 1500/1.079 = 110$ mm — order a 105 mm lens. Then check the consequence: $p_{\text{obj}} = 9.27/0.079 = 117 \mu\text{m}$, so a 1 mm feature spans only 8.5 pixels. That is marginal, and it is exactly the check people skip.

Never trust the formula for the final scale factor. The answer depends on where “1 metre” is measured from. Object-to-principal-plane gives $m = 0.111$ and a 213 mm field; object-to-sensor (total conjugate) gives $m = 0.127$ and a 187 mm field — a 14% difference from the same stated distance. Real lenses compound this: principal plane locations are undisclosed, internally-focusing designs change effective focal length with focus (breathing), and engraved focal lengths are nominal.

Use the formulae to *design* the setup. Then *calibrate*: place a ruler or dot target exactly in the measurement plane with focus untouched, image it, and divide known millimetres by measured pixels. This absorbs focal length error, principal plane uncertainty and breathing in one step, and is good to a fraction of a percent. Stereo PIV and stereo DIC require a full target calibration in any case, which also solves lens distortion.

4.2 Sensor-limited optical resolution

The sampling ceiling is one line pair per two pixels:

$$\nu_{\text{Nyquist}} = \frac{1}{2p} = 53.9 \text{ lp/mm} \quad (27.0 \text{ lp/mm binned}) \quad (5)$$

This is a modest demand — most competent lenses deliver usable MTF at 54 lp/mm. On a large-pixel high-speed camera the *sensor*, not the lens, is almost always the limiting element. Spend the optical budget on aperture speed rather than MTF, because photons are the scarce resource. Note also that a colour (Bayer) sensor delivers roughly 70–80% of the monochrome luminance resolution and considerably less chroma resolution; monochrome is correct for PIV, DIC and any quantitative work.

5 Focal length versus working distance

Detail per pixel is governed by magnification and nothing else, and both focal length and distance enter only through their ratio. This has a consequence people find surprising: **doubling the focal length and doubling the distance cancel exactly.** In Table 4, 90 mm at 1.0 m and 180 mm at 2.0 m give identical magnification, identical object-space pixel and identical field of view.

So why choose the long lens from far away? Three reasons, and only one is about sharpness.

5.1 Perspective — the real reason

Perspective is set by *distance alone*, never by focal length. Close in, near parts of a subject are proportionally much closer to the lens than far parts, so they image larger. Far away, the projection approaches orthographic. For measurement this is decisive: a close, wide setup introduces parallax error on anything with depth, and in PIV it means particles at the front of the light sheet carry a different scale factor than particles at the back. Backing off with a long lens is the correct instinct.

5.2 Working distance is often imposed

You cannot place a lens 200 mm from a flame, a detonation, or inside a tunnel wall. Long focal length is how standoff is purchased.

5.3 Depth of field follows magnification, not focal length

At equal magnification and equal f-number, a 90 mm at 1 m and a 180 mm at 2 m give essentially the same depth of field. The long lens costs nothing in depth. What costs depth is *more magnification* — and it costs quadratically. See Section 10.

Two limits on moving closer. First, minimum focus distance: a 180 mm lens that will not focus nearer than 1.5 m caps you at $m \approx 0.13$, and going further needs a macro lens, extension tubes or a bellows. Second, effective aperture grows with magnification as $N_w = N(1 + m)$, so at 1:1 an f/8 setting behaves like f/16 — one more stop of diffraction blur and two stops less light. At PIV magnifications ($m \approx 0.1$) this is a 10% correction.

6 The f-number

The f-number is focal length divided by entrance pupil diameter:

$$N = \frac{f}{D} \quad (6)$$

The notation “f/11” is a literal instruction: divide the focal length by 11 to obtain the pupil diameter. A 100 mm lens at f/2.8 has a 35.7 mm pupil; at f/13, a 7.7 mm pupil. A larger N therefore means a *smaller* hole. The standard series (1.4, 2, 2.8, 4, 5.6, 8, 11, 16, 22) steps by $\sqrt{2}$ so that each stop halves the pupil *area*, and hence the light. At finite conjugates, use the working f-number:

$$N_w = N(1 + m) \quad (7)$$

Aperture does three jobs simultaneously:

Effect	Scaling	Stopping down...
Illuminance on the sensor	$\propto 1/N^2$	makes it darker
Depth of field	$\propto N$	makes it deeper
Diffraction blur	$\propto N$	makes it blurrier

In ordinary photography these fight and you compromise. PIV is unusual: the third row, normally a defect, is the property being deliberately exploited.

7 Diffraction and the Airy disc

Light is a wave, and a finite aperture truncates the incoming wavefront. A truncated wave spreads — the narrower the truncation, the wider the spread, in the same way that ocean swell fans out more through a narrow harbour mouth than a wide one. A perfect lens therefore cannot image a point source as a point. It produces an *Airy pattern*: a bright central disc surrounded by faint rings.

The intensity distribution is governed by a first-order Bessel function J_1 , whose first zero lies at 3.8317. Dividing by π gives 1.22, the constant in the angular radius to the first dark ring:

$$\theta = \frac{1.22 \lambda}{D} \quad (8)$$

The pupil diameter D sits in the denominator: **a smaller hole produces a larger spot**. Equivalently, since the numerical aperture is $\text{NA} = 1/(2N)$, the f-number really describes the convergence half-angle of the light cone — a wide cone carries information from many angles and can construct a tight focus, while a narrow cone lacks the angular content to do so.

Converting the angle to a size at the image plane, which sits $s' = f(1 + m)$ behind the lens, and doubling for a diameter:

$$d_{\text{diff}} = 2.44 \lambda N(1 + m) \quad (9)$$

The constant $2.44 = 2 \times 1.22$ is fixed by the aperture being circular; a square aperture gives a different constant. It is not empirical — it follows from the geometry of a round hole, derived by George Biddell Airy in 1835 [14, 10].

8 Particle image diameter

The recorded image of a tracer particle combines two independent blurs: the geometrically projected particle and the diffraction pattern of the lens. Each behaves approximately as a Gaussian point-spread function, and convolving two Gaussians produces a third whose width is the root-sum-square of the originals [1, 2]:

$$d_\tau = \sqrt{(m d_p)^2 + d_{\text{diff}}^2} \quad (10)$$

where d_p is the physical seed particle diameter. Evaluated for the reference system at f/13, with 10 μm seed at $m = 0.111$ and $\lambda = 0.527 \mu\text{m}$:

$$m d_p = 0.111 \times 10 = 1.11 \mu\text{m} \quad (11)$$

$$d_{\text{diff}} = 2.44 \times 0.527 \times 13 \times 1.111 = 18.6 \mu\text{m} \quad (12)$$

$$d_\tau = \sqrt{1.11^2 + 18.6^2} = 18.6 \mu\text{m} \quad (13)$$

The geometric contribution is 0.03%. At low magnification you are photographing the Airy disc of your lens, not the particle. This has a liberating consequence — the geometric term drops out, and the aperture required for a target particle image size can be solved directly:

$$N \approx \frac{d_\tau}{2.44 \lambda (1 + m)} \quad \text{valid when } m d_p \ll d_{\text{diff}} \quad (14)$$

For a two-pixel target (18.54 μm): $N = 18.54 / (2.44 \times 0.527 \times 1.111) = \mathbf{f/13}$.

f-number	d_τ	Pixels	Assessment	Light vs f/5.6
f/2.8	4.0 μm	0.43	Severe peak locking	4.0 $\times$ more
f/5.6	8.0 μm	0.86	Peak locking	1 $\times$
f/8	11.4 μm	1.23	Marginal	2.0 $\times$ less
f/11	15.7 μm	1.69	Usable	3.9 $\times$ less
f/13	18.6 μm	2.00	Target	5.4$\times$ less
f/16	22.9 μm	2.46	Good	8.2 $\times$ less
f/22	31.4 μm	3.39	Acceptable but dark	15.4 $\times$ less

Table 5: Particle image diameter versus aperture at $m = 0.111$, $\lambda = 527 \text{ nm}$, $p = 9.27 \mu\text{m}$.

Particle image size is an optics problem, not a seeding problem. To reach 18.5 μm through the geometric term alone you would need 167 μm particles, whose Stokes number is far too high to follow the flow — they would measure their own inertia. Aperture is the only practical control.

9 Peak locking and the 2–3 pixel rule

PIV does not measure displacement to the nearest pixel. Cross-correlation locates a peak pixel, then a sub-pixel estimator — usually a three-point Gaussian fit across the peak and its two neighbours — refines the position to a fraction of a pixel, typically 0.05–0.1 px [7]. That fit requires the correlation peak to be sampled by more than one pixel, and the peak width derives from the particle image width.

- **Below about 1.5 px:** the correlation peak collapses onto a single pixel, the sub-pixel estimator degenerates, and displacement estimates are biased toward integer values. This is *peak locking*. It is diagnosable by plotting a histogram of measured displacements — spikes at 0, 1, 2 px with valleys between, instead of a smooth distribution. It corrupts turbulence

statistics, biases RMS values, and injects spurious energy into spectra [8]. It is the most common way a PIV dataset is quietly wrong.

- **2–3 px:** the working range. Enough samples across the peak for a stable fit, compact enough to preserve spatial resolution.
- **Above about 4 px:** the same photons spread over more pixels so per-particle SNR falls, and at useful seeding densities neighbouring particle images begin to overlap.

Prasad et al. [3] identified the two competing error mechanisms directly: mean bias error dominates when the particle image is small relative to the pixel, random error dominates when it is large. The optimum sits between.

9.1 Choosing Δt

The quarter rule [4]: particle displacement between the two exposures should be roughly 5–8 pixels, and must not exceed one quarter of the interrogation window:

$$\Delta t \approx \frac{(5-8) p_{\text{obj}}}{U} \quad (15)$$

At $p_{\text{obj}} = 83.4 \mu\text{m}$, eight pixels corresponds to 0.67 mm of real displacement.

Flow speed U	Required Δt
1 m/s	667 μs
5 m/s	133 μs
10 m/s	67 μs
50 m/s	13.3 μs
200 m/s	3.3 μs

With a 364 ns straddle floor, the camera is never the limiting factor on Δt . Laser pulse-to-pulse jitter and the finite (~ 100 ns) pulse width bite first, somewhere around $\Delta t = 1 \mu\text{s}$.

10 Depth of field

$$\text{DOF} \approx \frac{2 N c (1 + m)}{m^2} \quad (16)$$

where c is the circle of confusion; for PIV take $c = 2$ pixels = $18.54 \mu\text{m}$. The m^2 in the denominator is the important term. Doubling magnification quarters the depth of field. This — not focal length — is the price of “getting closer for more detail”.

Magnification	Object pixel	DOF at f/11	DOF at f/13
$m = 0.10$	93 μm	44.9 mm	53.0 mm
$m = 0.111$	83 μm	36.7 mm	43.4 mm
$m = 0.25$	37 μm	8.2 mm	9.6 mm
$m = 0.5$	19 μm	2.4 mm	2.9 mm
$m = 1.0$ (macro)	9.3 μm	0.82 mm	0.96 mm

At the low magnifications typical of planar PIV, a 1–2 mm light sheet sits inside the depth of field with enormous margin. In that regime aperture is chosen entirely by particle image size (Section 8) and depth of field never constrains the design. At micro-PIV magnifications the relationship inverts completely and depth of field becomes the dominant constraint.

11 Vignetting and corner falloff

Corner darkening at wide apertures has three distinct causes, worth separating because only one is fixed by stopping down.

11.1 Natural vignetting — the $\cos^4$ law

An off-axis pixel sees the exit pupil foreshortened into an ellipse ($\cos$), further away so subtending less solid angle ($\cos^2$), and receives light obliquely so it spreads over more area ($\cos$). The product gives

$$\frac{E(\theta)}{E(0)} = \cos^4 \theta \quad (17)$$

For a 28.2 mm diagonal sensor with a 25 mm lens the corner sits at $\theta = 29.4^\circ$, and $\cos^4(29.4^\circ) = 0.57$ — about 0.8 stop. This depends on focal length and format, not aperture [12].

11.2 Mechanical vignetting

At wide apertures the light cone reaching a corner is both wide and steeply tilted, so parts of it are clipped by the barrel, filter threads, retaining rings or hood. The effective corner pupil becomes a “cat’s eye” rather than a circle. Stopping down narrows the cone until nothing clips, which is why this component largely disappears by $f/4$. **This is the mechanism that makes corner falloff specifically an $f/1.4$ problem.**

11.3 Pixel angular response

A front-illuminated pixel places the photodiode beneath several microns of metal wiring, so steeply-arriving light strikes the sidewalls and is lost — the drinking-straw effect. The marginal ray half-angle is

$$u = \arcsin\left(\frac{1}{2N}\right) \quad (18)$$

giving 21° at $f/1.4$ against 7° at $f/4$; the chief ray angle at the corner of a non-telecentric lens adds a further $15\text{--}25^\circ$. Back-side illumination moves the wiring behind the photodiode, widening the acceptance angle considerably [16]. This is a principal reason BSI sensors hold up at fast apertures.

A related effect: microlens arrays are shifted progressively outward toward the sensor edge, pre-aimed at an assumed exit pupil distance. Lenses whose exit pupil sits much closer than that design assumption — typically short-flange retrofocus wide-angles — reintroduce the mismatch.

Correction. Falloff is a multiplicative shading field, so it degrades corner SNR and can bias correlation. Record a flat-field reference (uniformly lit white card at the same lens, aperture and focus) and divide. At PIV apertures of $f/11\text{--}f/16$ the mechanical component has already vanished, so this is rarely a problem in practice — it matters most in high-speed cinematography and low-light work shot wide open.

12 Worked design: 1 kHz time-resolved PIV

Pairing this camera with a 1 kHz dual-cavity laser puts the system in an unusual and favourable regime: **the laser is the bottleneck, not the camera.** The camera’s surplus capacity can therefore be spent on resolution instead of speed.

A convenient coincidence: EMVA 1288 sensitivity figures for this sensor family are measured at 532 nm [15], essentially the Nd:YLF second-harmonic line at 527 nm. The quoted 85.4% quantum efficiency therefore applies at the working wavelength without extrapolation. Use the monochrome variant — a Bayer filter discards roughly 30% of the signal for no benefit in a monochromatic, quantitative measurement.

Quantity	Value	Reasoning
Vector field rate	1 kHz	One pulse pair per cavity cycle
Required frame rate	2,000 fps	Straddling: two frames per pair
Camera capability	5,840 fps	Running at 34% of capacity
Sensor use	Full 2560×1664	No cropping or binning needed
Vectors (32 px, 50% overlap)	$160 \times 104 = 16,640$	2.7 mm windows, 1.3 mm spacing
Vectors (24 px, 50% overlap)	$213 \times 138 = 29,394$	2.0 mm windows, 1.0 mm spacing
Record time, 256 GB	20.0 s	$40,062 \text{ frames} \div 2,000 \text{ fps}$
Vector fields per run	$\approx 20,000$	Converged statistics
Aperture	f/13	2 px particle images
Depth of field	43 mm	Light sheet fits with margin

Table 6: Design summary for 1 kHz time-resolved PIV.

12.1 Is the measurement genuinely time-resolved?

Sampling at 1 kHz resolves flow phenomena to a Nyquist limit of 500 Hz, and confidently to perhaps 100 Hz. Test your case against the dominant frequency. A 20 mm cylinder in a 10 m/s stream sheds at $St U/D \approx 0.2 \times 10/0.02 = 100 \text{ Hz}$, giving ten samples per shedding cycle — fully resolved. A 5 mm feature at 50 m/s sheds at 2 kHz and is not resolved; that needs a faster laser, not a faster camera.

Well suited (time-resolved)	Ensemble or phase-locked only
Water channel and tunnel work, 0.1–2 m/s	Transonic and supersonic flow
Bluff-body wakes, vortex shedding	Shock–boundary-layer interaction
Jets, mixing layers, shear layers	Detonation and blast
Low-speed wind tunnel to ~ 20 –30 m/s	Combustion instability above $\sim 500 \text{ Hz}$
Channel and boundary-layer turbulence	High-Re turbulence spectra
Cardiovascular phantoms, biological locomotion	
Stirred tanks, mixing vessels	

For the right-hand column the laser repetition rate is irrelevant — you accumulate statistically independent realisations instead, and here a long RAM buffer and multi-partition recording make the camera very effective at collecting thousands of samples.

12.2 Recovering the light lost to f/13

1. **Pulse energy.** The correct answer. A 1 kHz dual-cavity Nd:YLF typically delivers 10–20 mJ per pulse per cavity, sized for exactly this.
2. **Thinner light sheet and better seeding.** Concentrating the same energy into a thinner sheet raises irradiance per particle, free within the depth of field budget.
3. **Slight defocus instead of stopping down.** A last resort. It does enlarge particle images at no light cost, but defocus blur is neither symmetric nor uniform across the field, varies through the sheet thickness, and distorts the correlation peak shape in ways that bias the sub-pixel estimator. Stopping down is predictable and identical everywhere in the frame.

13 Formula sheet

14 Notation

References

- [1] M. Raffel, C. E. Willert, F. Scarano, C. J. Kähler, S. T. Wereley and J. Kompenhans, *Particle Image Velocimetry: A Practical Guide*, 3rd ed. Springer, 2018.
- [2] R. J. Adrian and J. Westerweel, *Particle Image Velocimetry*. Cambridge University Press, 2011.

Quantity	Expression	Notes
Frame rate	$\text{fps} = \text{throughput}/(H \times V)$	Throughput in px/s
Frame size	$\text{bytes} = HVb/8$	b = bit depth
Magnification	$m = f/(s - f)$	s to front principal plane
Field of view	$\text{FOV} = \text{sensor dim.}/m$	Per axis
Object-space pixel	$p_{\text{obj}} = p/m$	The measurement quantum
Lens selection	$f = ms/(1 + m)$	From FOV and standoff
Nyquist limit	$\nu = 1/(2p)$	Line pairs per mm
f-number	$N = f/D$	D = entrance pupil
Working f-number	$N_w = N(1 + m)$	Finite conjugates
Numerical aperture	$\text{NA} = 1/(2N)$	Convergence half-angle
Airy disc diameter	$d_{\text{diff}} = 2.44\lambda N(1 + m)$	To first dark ring
Particle image	$d_\tau = \sqrt{(md_p)^2 + d_{\text{diff}}^2}$	Target 2–3 px
Required aperture	$N \approx d_\tau/[2.44\lambda(1 + m)]$	When $md_p \ll d_{\text{diff}}$
Depth of field	$\text{DOF} \approx 2Nc(1 + m)/m^2$	$c = 2$ px for PIV
Natural vignetting	$E(\theta)/E(0) = \cos^4 \theta$	Off-axis illuminance
Marginal ray angle	$u = \arcsin(1/2N)$	Relevant to sensor CRA
PIV time step	$\Delta t \approx (5\text{--}8)p_{\text{obj}}/U$	Quarter-rule bounded
Vector count	$(H/\text{step}) \times (V/\text{step})$	$\text{step} = \text{window} \times (1 - \text{overlap})$

Symbol	Quantity	Units
p	Sensor pixel pitch	μm
p_{obj}	Object-space pixel (scale factor)	μm or mm/px
f	Focal length	mm
s, s'	Object and image distance	mm
m	Magnification	—
N	f-number	—
N_w	Working f-number	—
D	Entrance pupil diameter	mm
λ	Wavelength	μm or nm
d_p	Physical seed particle diameter	μm
d_{diff}	Airy disc diameter	μm
d_τ	Recorded particle image diameter	μm
c	Circle of confusion	μm
θ	Off-axis field angle	degrees
Δt	Inter-frame (pulse separation) time	μs
U	Flow velocity	m/s
St	Strouhal number	—

- [3] A. K. Prasad, R. J. Adrian, C. C. Landreth and P. W. Offutt, “Effect of resolution on the speed and accuracy of particle image velocimetry interrogation,” *Experiments in Fluids*, vol. 13, pp. 105–116, 1992. DOI: 10.1007/BF00218156.
- [4] R. D. Keane and R. J. Adrian, “Optimization of particle image velocimeters. Part I: Double pulsed systems,” *Measurement Science and Technology*, vol. 1, pp. 1202–1215, 1990.
- [5] R. D. Keane and R. J. Adrian, “Theory of cross-correlation analysis of PIV images,” *Applied Scientific Research*, vol. 49, pp. 191–215, 1992.
- [6] R. J. Adrian, “Particle-imaging techniques for experimental fluid mechanics,” *Annual Review of Fluid Mechanics*, vol. 23, pp. 261–304, 1991.
- [7] J. Westerweel, “Fundamentals of digital particle image velocimetry,” *Measurement Science and Technology*, vol. 8, pp. 1379–1392, 1997.

- [8] J. Nogueira, A. Lecuona and P. A. Rodríguez, “Identification of a new source of peak locking, analysis and its removal in conventional and super-resolution PIV techniques,” *Experiments in Fluids*, vol. 30, pp. 309–316, 2001.
- [9] R. J. Adrian, “Twenty years of particle image velocimetry,” *Experiments in Fluids*, vol. 39, pp. 159–169, 2005.
- [10] M. Born and E. Wolf, *Principles of Optics*, 7th ed. Cambridge University Press, 1999. Ch. 8 for the Airy pattern.
- [11] W. J. Smith, *Modern Optical Engineering*, 4th ed. McGraw-Hill, 2008.
- [12] S. F. Ray, *Applied Photographic Optics*, 3rd ed. Focal Press, 2002.
- [13] J. W. Goodman, *Introduction to Fourier Optics*, 4th ed. W. H. Freeman, 2017.
- [14] G. B. Airy, “On the diffraction of an object-glass with circular aperture,” *Transactions of the Cambridge Philosophical Society*, vol. 5, pp. 283–291, 1835.
- [15] European Machine Vision Association, *EMVA Standard 1288: Standard for Characterization of Image Sensors and Cameras*, Release 3.1, 2016.
- [16] E. R. Fossum and D. B. Hondongwa, “A review of the pinned photodiode for CCD and CMOS image sensors,” *IEEE Journal of the Electron Devices Society*, vol. 2, no. 3, pp. 33–43, 2014.
- [17] Vision Research (Ametek), *Phantom T4040 / T2540 Datasheet*. phantomhighspeed.com. Consult the revision matching your camera’s serial number; specifications have changed between revisions.
- [18] Vision Research (Ametek), *Phantom Ultrahigh-Speed Camera Manual*. Covers ISO 12232 SAT sensitivity methodology, black reference procedure and programmable I/O architecture.